Today

- Finite square well
- Delta function potential
- Reflection and transmission
- The simple harmonic oscillator

2.3 Finite square well

Question 2.3.1: Considering a finite square well of the same width as the infinite square well with $V(x) = 0$ for $-\frac{a}{2} \leq x \leq \frac{a}{2}$ and $V(x) = V_0$ for $|x| > \frac{a}{2}$, which of the following are true about bound states in the finite square well? (A). The wavefunction is nonzero outside the well. (B). The ground state is symmetric with no nodes. (C). $\psi_n(x)$ has the same number of nodes as the nodes in the interior for $\psi_n(x)$ of the infinite square well. (D). E_n is smaller than E_n of the infinite square well. (E). There are a finite number of bound states.

Now let us consider a particle of mass m in a potential well of finite depth,

$$V(x) = \begin{cases} -V_0, & -a \leq x \leq a \\ 0, & |x| > a. \end{cases}$$

The potential is shown in the figure below.

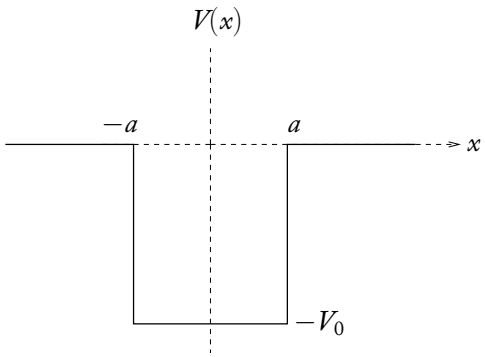


Figure 2.3.1: Finite square well.

Our interest here is to find the stationary bound states. Because the potential is piecewise and

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discontinuous, let us call the region on the left where $x < -a$ region *I*, that with in the well $a \leq x \leq a$ region *II*, and the region where $x > a$ region *III*. The Schrödinger equation in region *I* is

$$\psi_I''(x) - \kappa^2 \psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^1}}, \tag{2.3.1}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.3.2}$$

The Schrödinger equation and the general solution in region *III* are exactly the same as in region *I*. The only difference is that the wavefunction in region *III* must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{III}(x) = Ae^{-\kappa x}. \tag{2.3.3}$$

The Schrödinger equation in region *II* is

$$\psi_{II}''(x) + k^2 \psi_{II}(x) = 0, \qquad k \equiv \sqrt{\frac{2m(V_0 + E)}{\hbar^1}}, \tag{2.3.4}$$

Because we can choose the solutions to be real and because we know that the bound states will be either even or odd, it is convenient to choose the general solution as a linear combination of sin and cos rather than complex exponentials, and we write

$$\psi_{II}(x) = C \sin kx + D \cos kx. \tag{2.3.5}$$

$C = 0$ and $D \neq 0$ correspond to even solutions, while $D = 0$ and $C \neq 0$. Let us investigate each case separately.

2.3.1 Even bound states

Even bound states with $\psi(-x) = \psi(x)$ pick out

$$\psi_{II}(x) = D \cos kx. \tag{2.3.6}$$

Because the potential has only a finite discontinuity at $x = \pm a$, both the wavefunction and its first derivative must be continuous at $x = \pm a$. Continuity of $\psi(x)$ at $x = a$, $\psi_{II}(a) = \psi_{III}(a)$,

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imposes

$$D\cos ka = Ae^{-\kappa a}. \tag{2.3.7}$$

Continuity of $\psi'(x)$ at $x = a$, $\psi'_{II}(a) = \psi'_{III}(a)$, imposes

$$-kD\sin ka = -\kappa Ae^{-\kappa a}. \tag{2.3.8}$$

Imposing continuity of $\psi(x)$ and $\psi'(x)$ at $x = -a$ does not add anything new; it reproduces the same constraints. Dividing the above two, we eliminate the constant coefficients, and we have

$$k\tan ka = \kappa. \tag{2.3.9}$$

Observe that given a and V_0, this is a constraint on the allowed energies of the bound states. However, this is a transcendental equation that does not have analytic solutions. We can solve it numerically or graphically as we discuss below. Using the definitions of k and κ and rewriting, $\tan ka = \frac{\kappa}{k}$, or $\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}}\right) = \sqrt{\frac{-E}{V_0+E}} = \sqrt{\frac{-E-V_0+V_0}{V_0+E}} = \sqrt{-1+\frac{V_0}{V_0+E}}$; i.e.,

$$\tan\left(\sqrt{\frac{2m(V_0+E)a^2}{\hbar^1}}\right) = \sqrt{\frac{V_0}{V_0+E}-1}. \tag{2.3.10}$$

Defining new variables

$$z \equiv \sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}}, \qquad z_0 \equiv \sqrt{\frac{2mV_0a^2}{\hbar^2}}, \tag{2.3.11}$$

the constraint becomes

$$\tan z = \sqrt{(\frac{z_0}{z})^2-1}. \tag{2.3.12}$$

The solutions can be obtained by plotting each side of (2.3.12) and finding the intersection points as shown in the figure below; i.e. the values of z for which both sides are equal.

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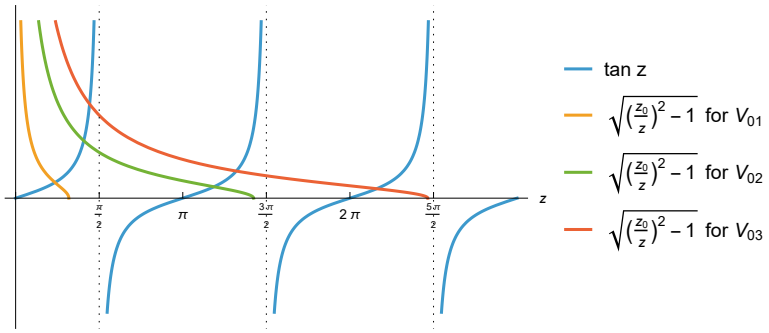


Figure 2.3.2: Graphic solution for the even bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential V_{01} produces only one bound state, since there is only one intersection point, V_{02} produces two bound state, and V_{03} produces three bound state. We also see that any finite well in one dimension supports at least one bound state.

2.3.2 Odd bound states

For the odd states,

$$\psi_{II}(x) = C \sin kx. \tag{2.3.13}$$

Continuity of $\psi(x)$ at $x = -a$, $\psi_{II}(-a) = \psi_I(-a)$ imposes

$$C \sin ka = Be^{-\kappa a}. \tag{2.3.14}$$

Continuity of $\psi'(x)$ at $x = -a$, $\psi'_{II}(-a) = \psi'_I(-a)$, imposes

$$kC \cos ka = -\kappa Be^{-\kappa a}. \tag{2.3.15}$$

Dividing the above two,

$$k \cot ka = -\kappa. \tag{2.3.16}$$

Rewriting this in terms of z and z_0 as in the even solutions, the constraint becomes

$$\cot z = -\sqrt{(\frac{z_0}{z})^2 - 1}. \tag{2.3.17}$$

This is also a transcendental equation which we solve using graphical method as shown in the figure below

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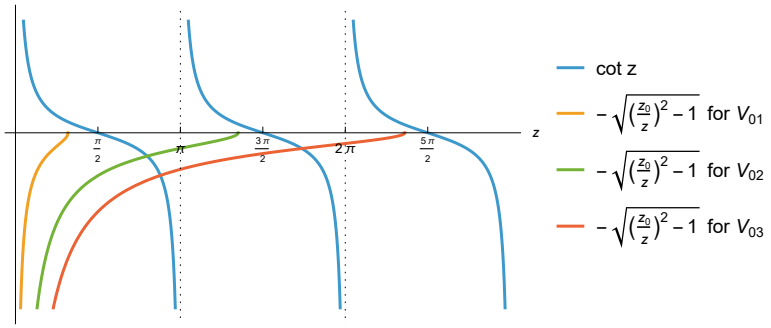


Figure 2.3.3: Graphic solution for the odd bound states of the finite square well with depth $V_{01} < V_{02} < V_{03}$.

In the figure above, the potential V_{01} does not support an odd bound state, since there is no intersection point, V_{02} produce only one bound state, and V_{03} produces two bound state. It means that a finite well potential supports at lease one even state, the ground state, and it may not support any of the excited states.

2.3.3 Wide and deep well

> **Question 2.3.2:** A wider well of the same depth supports (A). the same number of bound states. (B). larger number of bound states. (C). smaller number of bound states. (D). the same or less number of bound states. (E). the same or more number of bound states.

Now suppose the width and depth of the well are such that $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} >> 1$. In this case, the curve $\sqrt{(\frac{z_0}{z})^2-1}$ for the odd case begins high above the $z-$axis and intersects $\tan z$ at its asymptotic values of $\frac{\pi}{2}$, $\frac{3\pi}{2}$, etc, as shown in the figure below.

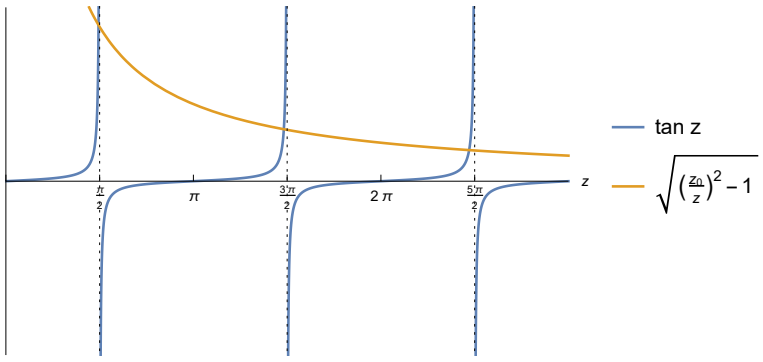


Figure 2.3.4: Graphic solution for the even bound states of a wide and deep finite square well.

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Therefore, the intersection points for the odd states are at $z_n \approx \frac{n\pi}{2}$, n=1,3,5,... Similarly for the odd solutions, $z_n \approx \frac{n\pi}{2}$, $n = 2, 4, 6, \ldots$ Combining both the even and odd cases,

$$z_n \approx \frac{n\pi}{2}, \quad n = 1, 2, 3, \ldots \tag{2.3.18}$$

Using the definition of z, $\sqrt{\frac{2m(V_0+E)a^2}{\hbar^2}} \approx \frac{n\pi}{2}$, and solving for E,

$$E_n = -V_0 + \frac{\hbar^2\pi^2 n^2}{8ma^2}, \quad n = 1, 2, 3, \ldots \tag{2.3.19}$$

Thus, the energy levels of the infinite square well of width $2a$ are reproduced for $E_n - (-V_0) = E_n + V_0$, since the ground state energy is shifted down and begins near $E = -V_0$ for the given potential.

2.3.4 Shallow and narrow well

We have already seen that any shallow or narrow well in one-dimension supports at least one bound state. But we need at least $z_0 = \sqrt{\frac{2mV_0a^2}{\hbar^2}} \geq \frac{\pi}{2}$ to support any odd state, as seen in Figure 2.3.3.

Question 2.3.3: Why is the ground state energy of the quantum oscillator (or any bound particle in a potential well) not zero, or why is the particle never at rest?

Question 2.3.4: Why are the energy levels of the finite square well smaller than the corresponding energy levels of the infinite square well?

2.4 Delta function potential

Consider a particle of mass m in a delta function potential well,

$$V(x) = -\alpha\delta(x), \tag{2.4.1}$$

where α is a positive constant. A schematic figure of the potential is shown below.

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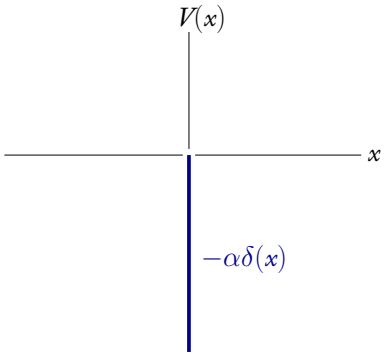


Figure 2.4.1: Delta function potential well.

To construct any bound state wavefunctions and the corresponding energies, we solve the Schrödinger equation on the left and the right sides of the potential and impose the continuity of $\psi(x)$ and the discontinuity of $\psi'(x)$ across it. Note that now that the potential has infinite discontinuity at the origin, $\psi'(x)$ is not continuous. Denoting the region $x < 0$ by I and the region $x > 0$ by II, the Schrödinger equation in region I is

$$\psi_I''(x) - \kappa^2 \psi_I(x) = 0, \qquad \kappa \equiv \sqrt{-\frac{2mE}{\hbar^2}}, \tag{2.4.2}$$

where $E < 0$ for a bound state. The general solution is $\psi_I(x) = Ae^{-\kappa x} + Be^{\kappa x}$. But because the wavefunction has to vanish at $x = -\infty$, in order for it to be normalizable, we set $A = 0$, and we have

$$\psi_I(x) = Be^{\kappa x}. \tag{2.4.3}$$

The Schrödinger equation and the general solution in region II are exactly the same as in region I. The only difference is that the wavefunction in region II must vanish at $x = \infty$, which sets $B = 0$, and we have

$$\psi_{II}(x) = Ae^{-\kappa x}. \tag{2.4.4}$$

Continuity of the wavefunction across $x = 0$ imposes

$$A = B. \tag{2.4.5}$$

The discontinuity of $\psi'(x)$ across $x = 0$ follows from

$$\psi_{II}'(0) - \psi_I'(0) \;=\; \lim_{\epsilon \to 0} \int_{-\epsilon}^{\epsilon} \psi''(x)dx = \lim_{\epsilon \to 0} \int_{-\epsilon}^{\epsilon} \frac{2m}{\hbar^2}(V(x) - E)\psi(x)dx$$

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$$= -\lim_{\epsilon\to 0}\frac{2m}{\hbar^2}\int_{-\epsilon}^{+\epsilon}\alpha\delta(x)\psi(x)dx = -\frac{2m}{\hbar^2}\alpha\psi(0) = -\frac{2m}{\hbar^2}\alpha A, \tag{2.4.6}$$

On the other hand, taking the derivatives directly, $\psi'_{II}(0) - \psi'_{I}(0) = -\kappa A - \kappa B$. Therefore,

$$\kappa A + \kappa B = \frac{2m}{\hbar^2}\alpha A. \tag{2.4.7}$$

Solving (2.4.5) and (2.4.7), we have

$$\kappa = \frac{m\alpha}{\hbar^2}, \tag{2.4.8}$$

which using the definition of κ gives

$$E = -\frac{m\alpha^2}{2\hbar^2}. \tag{2.4.9}$$

Because there is only one solution, the delta function potential well supports one and only one bound state. Because $A = B$, the wavefunction decays symmetrically and exponentially on both sides. Writing $\psi_I(x) = Ae^{\kappa x}$ and $\psi_{II}(x) = Ae^{\kappa x}$, and normalizing, $1 = \int_{-\infty}^{0} A^2 e^{2\kappa x} + \int_{0}^{\infty} A^2 e^{-2\kappa x} = A^2\frac{1}{2\kappa} + A^2\frac{1}{2\kappa} = \frac{A^2}{\kappa}$, which gives $A = \sqrt{\kappa}$. Therefore, the normalized wavefunction is

$$\psi(x) = \begin{cases} \sqrt{\kappa}e^{-\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{-\frac{m\alpha}{\hbar^2}x}, & x > 0 \\ \sqrt{\kappa}e^{\kappa x} = \sqrt{\frac{m\alpha}{\hbar^2}}e^{\frac{m\alpha}{\hbar^2}x}, & x < 0. \end{cases}$$

The energy level and the wavefunction are shown in the schematic figure below.

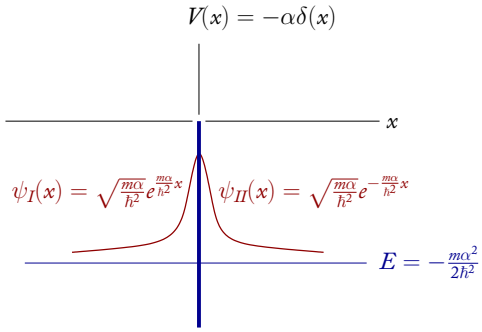


Figure 2.4.2: Energy and wavefunction of the delta function potential well.

2.5 Reflection and transmission

Now we want to consider the case of a particle of energy $E > 0$ approaching the well from the left of the finite square well potential whose bound states we have investigated. Now the

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particle has positive kinetic energy everywhere above the potential and it moves along from $x = -\infty$ to $x = \infty$.

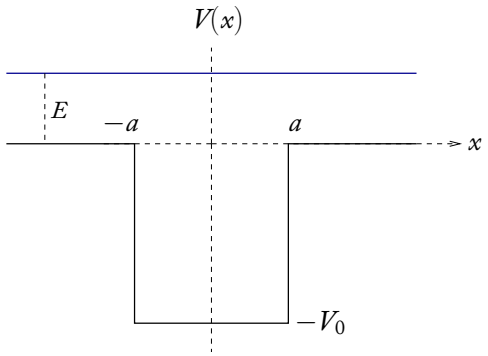


Figure 2.5.1: A particle with energy above an infinite square well.

Question 2.5.1: For a particle represented by a plane wave with energy $E > 0$ approaching the well from the left of an finite square well, (A). the wavefunction vanishes at $x = \pm\infty$. (B). the particle is partly reflected and partly transmitted at $x = -a$. (C). the particle is partly reflected and partly transmitted at $x = a$. (D). the particle is partly reflected and partly transmitted at all points it passes through. (E). the speed of the particle increases when it is between $x = -a$ and $x = a$. (F). the wavelength of the particle decreases when it is between $x = -a$ and $x = a$.

Question 2.5.2: The coefficient of transmission T should in general be defined as (A). $\frac{J_{tra}}{J_{inc}}$, (B). $\left|\frac{\psi_{tra}}{\psi_{inc}}\right|^2$, (C). Either one of the above.

Recalling that the probability current describes the particle current, the rate of particle flow though a given point, we should note that in order to compute the probability for reflection or transmission, we compare the corresponding currents. The coefficients of reflection R and transmission T are defined as

$$R = \frac{J_{ref}}{J_{inc}}, \qquad T = \frac{J_{tra}}{J_{inc}}, \tag{2.5.1}$$

where J_{inc} is the probability current for the incident wave, J_{ref} is the probability current for the reflected wave, and J_{tra} is the probability current for the transmitted wave. Observe that comparing the probability densities only does not provide a correct answer, unless the speed of the particle is the same in the corresponding regions.

Considering the finite potential well shown in the figure above, and denoting the region in $x < -a$ by I, that within $-a \leq x \leq a$ by II, and the region in $x > a$ by III, the solutions to the

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Schrödinger equation in each region can be written down immediately,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \tag{2.5.2}$$
$$\psi_{II}(x) = C\sin\kappa x + D\cos\kappa x, \tag{2.5.3}$$
$$\psi_{III}(x) = Fe^{ikx}, \tag{2.5.4}$$

where

$$\kappa \equiv \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa \equiv \sqrt{\frac{2m(E+V_0)}{\hbar^2}}, \tag{2.5.5}$$

A is the amplitude of the incident wave, B is the amplitude of the reflected wave, and F is the amplitude of the transmitted wave. There is no reflected wave that propagates to the left in region *III*. Applying the continuity of $\psi(x)$ and $\psi'(x)$ at $x = \pm a$ gives the following four equations,

$$Ae^{-ika} + Be^{ika} = -C\sin\kappa a + D\cos\kappa a, \tag{2.5.6}$$
$$ik\left(Ae^{-ika} - Be^{ika}\right) = \kappa\left(C\cos\kappa a + D\sin\kappa a\right), \tag{2.5.7}$$
$$C\sin\kappa a + D\cos\kappa a = Fe^{ika}, \tag{2.5.8}$$
$$\kappa\left(C\cos\kappa a - D\sin\kappa a\right) = ikFe^{ika} \tag{2.5.9}$$

We have four equation which we can solve for four variables B, C, D, and F in terms of the amplitude A of the incident wave. Our interest is in the amplitudes B and F for the reflected and the transmitted waves, and we write the solutions for B and F,

$$F = \frac{e^{-2ika}}{\cos(2\kappa a) - i\frac{k^2+\kappa^2}{2k\kappa}\sin(2\kappa a)}A \tag{2.5.10}$$

and

$$B = i\frac{\sin(2\kappa a)}{2k\kappa}(\kappa^2 - k^2)F. \tag{2.5.11}$$

The transmission coefficient is obtained using Ae^{ikx} for the incident wave and Fe^{ikx} for the transmitted wave, which gives

$$T = \frac{J_{tra}}{J_{inc}} = \left|\frac{F}{A}\right|^2 = \frac{1}{1 + \frac{V_0^2}{4E(E+V_0)}\sin^2\left(\frac{2a}{\hbar}\sqrt{2m(E+V_0)}\right)}, \tag{2.5.12}$$

and the transmission coefficient oscillates with the energy of the incident particle. More-

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over,

$$R = 1 - T. \tag{2.5.13}$$

The transmission coefficient oscillates with energy as shown in the figure below.

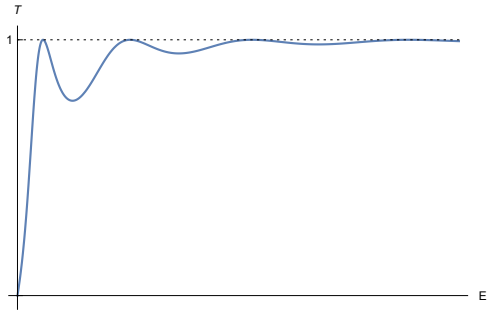
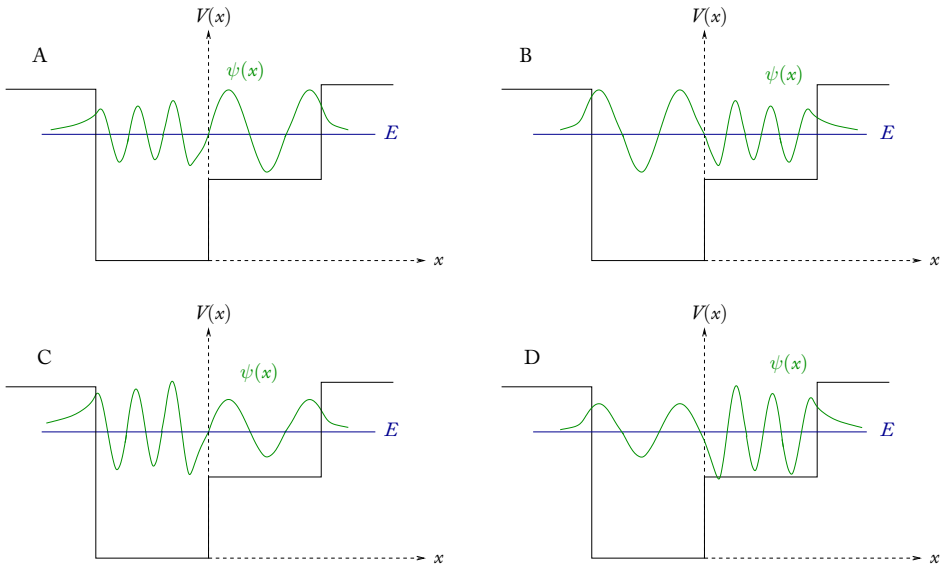


Figure 2.5.2: Transmission coefficient for a particle over a square potential well.

Question 2.5.3: Discuss why the transmission coefficient oscillates with the energy of the incident particle and what the peaks correspond to.

Question 2.5.4: Which one of the following best describes the wavefunction of a bound particle of energy E in the potential well $V(x)$?



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2.6 Step potential

Another example is a particle of energy $E > V_0$ approaching a step potential from the left in the $x < 0$ region with

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x > 0. \end{cases}$$

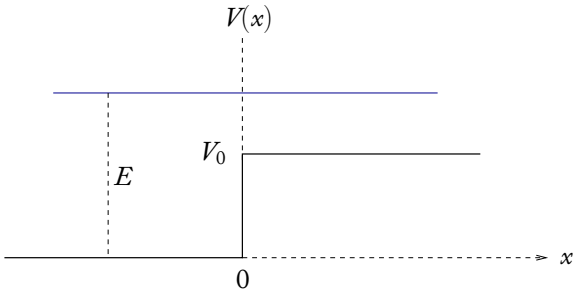


Figure 2.6.1: A particle with energy above a step potential.

Denoting the region $x < 0$ by I and that in $x > 0$ by II, and noting that there are both incident and reflected waves in region I while there is only transmitted wave in region II, we can immediately write the wavefunctions in each region,

$$\psi_I(x) = Ae^{ikx} + Be^{-ikx}, \qquad \psi_{II}(x) = Ce^{i\kappa x}, \tag{2.6.1}$$

where

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \kappa = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} \tag{2.6.2}$$

Matching $\psi(x)$ and $\psi'(x)$ across $x = 0$, we have

$$A + B = C, \qquad ik(A - B) = i\kappa C. \tag{2.6.3}$$

Solving these simultaneous equations for B and C in terms of A,

$$B = \frac{k - \kappa}{k + \kappa}A, \qquad C = \frac{2k}{k + \kappa}A. \tag{2.6.4}$$

Now that the speed of the particle is different in the two regions we must use the probability current to calculate the reflection and transmission coefficients. The probability currents for

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incident, reflected, and transmitted waves are

$$J_{inc} = \frac{\hbar k}{m}|A|^2, \quad J_{ref} = \frac{\hbar k}{m}|B|^2, \qquad J_{tra} = \frac{\hbar \kappa}{m}|C|^2. \tag{2.6.5}$$

The reflection and transmission coefficients are

$$\begin{aligned} R = \frac{J_{ref}}{J_{inc}} &= \left|\frac{B}{A}\right|^2 = \left(\frac{k-\kappa}{k+\kappa}\right)^2 = \left(\frac{\sqrt{E}-\sqrt{E-V_0}}{\sqrt{E}+\sqrt{E-V_0}}\right)^2 \\ T = \frac{J_{tra}}{J_{inc}} &= \frac{\kappa}{k}\left|\frac{C}{A}\right|^2 = \frac{4k\kappa}{(k+\kappa)^2} = \frac{4\sqrt{E}\sqrt{E-V_0}}{(\sqrt{E}+\sqrt{E-V_0})^2} \end{aligned} \tag{2.6.6}$$

Observe that

$$R + T = 1, \tag{2.6.7}$$

consistent with conservation of probability. The coefficients and their sum as a function of energy are shown in the figure below.

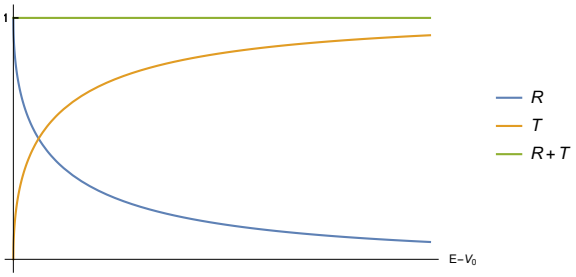


Figure 2.6.2: Reflection and transmission coefficients and their sum for a particle with energy above a step potential.

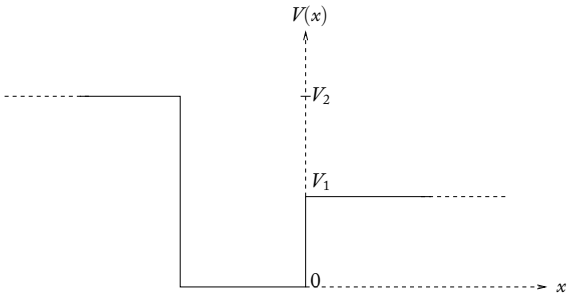
Question 2.6.1: Compare the reflection or transmission of the quantum particle discussed above with a classical particle.

Question 2.6.2: For $E < V_0$ in the step potential, (A). $B = A$, (B). $B = -A$, (C). $T = 1$, (D). $R = 1$,

Question 2.6.3: For $E << V_0$ in the step potential, (A). $B \to A$, (B). $B \to -A$, (C). $C \to A$, (D). $C \to -A$,

Question 2.6.4: Which of the following statements are true about the energies of a particle in the potential shown below?

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(A). The energies are discrete for $0 < E < V_1$. (B). The energies are discrete for $V_1 < E < V_2$. (C). The energies are discrete for $E > V_2$. (D). The energies are continuous for $0 < E < V_1$. (E). The energies are continuous for $V_1 < E < V_2$. (F). The energies are continuous for $E > V_2$.

2.7 The simple harmonic oscillator

Now we want to investigate a particle of mass m in the simple harmonic potential

$$V(x) = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2x^2. \tag{2.7.1}$$

Because the potential is time-independent, we start with solving the time-independent Schrödinger equation,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}m\omega^2x^2\psi(x) = E\psi(x). \tag{2.7.2}$$

We still need boundary conditions. Because the potential becomes infinite at $x = \pm\infty$, the particle cannot be found there, since it would need infinite energy. Alternatively, because the wavefunction needs to be normalizable, it has to vanish at $x = \pm\infty$. Therefore the boundary conditions are

$$\psi(-\infty) = 0, \qquad \psi(+\infty) = 0. \tag{2.7.3}$$

The wavefunction starts with $\psi(-\infty) = 0$ at $x = -\infty$, oscillates according to the Schrödinger equation in the interior, and falls back to $\psi(+\infty) = 0$ at $x = +\infty$. This equation is not easy to solve immediately. Therefore, we follow a number of steps.

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Step 1: Simplify notation with change of variable

The first step is to rewrite the equation in simpler form without too many variables floating around. To do that, we make a change of variable of the coordinate x and define

$$\xi = \sqrt{\frac{m\omega}{\hbar}}x. \tag{2.7.4}$$

We then have $\frac{d}{dx} = \frac{d\xi}{dx}\frac{d}{d\xi} = \sqrt{\frac{m\omega}{\hbar}}\frac{d}{d\xi}$ and $\frac{d^2}{dx^2} = \frac{m\omega}{\hbar}\frac{d^2}{d\xi^2}$. The Schrödinger equation then takes the form

$$\frac{d^2\psi(\xi)}{d\xi^2} = (\xi^2 - K)\psi(\xi), \tag{2.7.5}$$

where

$$K \equiv \frac{2E}{\hbar\omega}. \tag{2.7.6}$$

Step 2: Asymptotic behavior

If we have an approximate solution in some region of ξ, then we can write a solution ansatz that is a product of the approximate solution and an unknown function $H(\xi)$ that would allow to solve the general equation exactly. Hopefully, the equation for $H(\xi)$ is easier to solve. In particular, (2.7.5) is difficult because of the ξ^2 term. If we could have approximate solution that solves it at this order, then this term would be eliminated. Because the ξ^2 term dominates over the K term for large ξ, let us work in the large $\xi^2 >> K$ region, where the equation reduces to

$$\frac{d^2\psi(\xi)}{d\xi^2} \sim \xi^2\psi(\xi). \tag{2.7.7}$$

Now we can immediately write approximate solutions,

$$\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}, \qquad \psi(\xi) \sim e^{\frac{1}{2}\xi^2}. \tag{2.7.8}$$

Because a solution that goes like $e^{\frac{1}{2}\xi^2}$ at large ξ cannot be normalizable, we discard it and keep only $\psi(\xi) \sim e^{-\frac{1}{2}\xi^2}$. Note that this gives $\psi''(\xi) \sim \xi^2 e^{-\frac{1}{2}\xi^2} - e^{-\frac{1}{2}\xi^2}$ which approximately equals $\xi^2 e^{-\frac{1}{2}\xi^2}$, since it dominates over $-e^{-\frac{1}{2}\xi^2}$ for large ξ, and thus the approximate equation is approximately solved. That is all well, since we are not seeking exact solutions at this point. We then write an ansatz solution,

$$\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}; \tag{2.7.9}$$

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we want to solve for $H(\xi)$ exactly and produce the exact $\psi(\xi)$. Differentiating this,

$$\frac{d^2\psi(\xi)}{d\xi^2} = H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.10}$$

Putting (2.7.9) and (2.7.10) in (2.7.5), we have

$$H''(\xi)e^{-\frac{1}{2}\xi^2} - 2\xi H'(\xi)e^{-\frac{1}{2}\xi^2} + H(\xi)e^{-\frac{1}{2}\xi^2} + \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} = \cancel{\xi^2 H(\xi)e^{-\frac{1}{2}\xi^2}} - KH(\xi)e^{-\frac{1}{2}\xi^2}. \tag{2.7.11}$$

Observe that the ξ^2 terms cancel out, since the approximate solution $e^{-\frac{1}{2}\xi^2}$ solves the Schrödinger equation at leading order in large ξ.

Thus, the Schrödinger equation reduces to an equation for $H(\xi)$.

$$H''(\xi) - 2\xi H'(\xi) + (K-1)H(\xi) = 0. \tag{2.7.12}$$

This equation is called the Hermite equation.

Step 3: Method of power series

It is known in the theory of differential equations that if the equation is written with the coefficient of the highest order derivative set to one as in here, then the equation could be singular only at the singular points of the coefficients of the lower order derivative terms. The coefficients we have in (2.7.12) have no singularity and, therefore, $H(\xi)$ can have no singularities at finite ξ and a power series solution will have an infinite radius of convergence. Thus, we can write a power series solution of the form

$$H(\xi) = \sum_{n=0}^{\infty} a_n \xi^n. \tag{2.7.13}$$

The idea is then to rewrite (2.7.12) as $\sum[\ldots]_n\xi^n = 0$, with constant coefficients $[\ldots]_n$, which holds for all ξ only if each constant coefficient vanishes; i.e., $[\ldots]_n = 0$ for each n. Therefore, we seek to write each term in (2.7.12) as $\sum[\ldots]_n\xi^n = 0$ so that we can easily combine the terms together. First, $H'(\xi) = \sum_{n=0}^{\infty} n a_n \xi^{n-1}$. Because what we have in (2.7.12) is $-2\xi H'(\xi) = -2\sum_{n=0}^{\infty} n a_n \xi^n$, which has the desired form, we keep it as is. Next, taking the second derivative, $H''(\xi) = \sum_{n=0}^{\infty} n(n-1)a_n\xi^{n-2}$. Now this has ξ^{n-2} while we seek ξ^n so that we can combine the coefficients with the other terms. But the first two terms for $n = 0$ and $n = 1$ are zero, and we can start the sum with $n = 2$, $H''(\xi) = \sum_{n=2}^{\infty} n(n-1)a_n\xi^{n-2}$. Next we shift $n \to n+2$ and the sum starts at

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$n = 0$ as in the others, which we do by rewriting it as $H''(\xi) = \sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}\xi^n$, and we have expanded it in terms of ξ^n too. Putting these in (2.7.12), we have

$$\sum_{n=0}^{\infty}\left[(n+2)(n+1)a_{n+2} - (2n+1-K)a_n\right]\xi^n = 0. \tag{2.7.14}$$

This holds for all ξ only if each and every coefficient of ξ^n is identically zero, which gives the recursion relation

$$a_{n+2} = \frac{2n+1-K}{(n+2)(n+1)}a_n \tag{2.7.15}$$

Observe that there are only two independent coefficients, since starting with a_0, we recursively produce a_2, a_4, etc, which produces even solutions. If we start with a_1, we recursively obtain a_3, a_5, etc, which produces odd solutions.

Step 4: Infinite series produces a diverging function

For large n, we have

$$\frac{a_{n+2}}{a_n} \to \frac{2}{n} \tag{2.7.16}$$

But this is similar to the behavior in the series expansion of $\xi^m e^{\xi^2}$ for any m, which goes like e^{ξ^2} for large ξ. In other words, if we keep all infinite series of terms, then we would have $H(\xi) \sim e^{\xi^2}$ and $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2} \sim e^{\frac{1}{2}\xi^2}$, and the wavefunction would blow up at large ξ and would not be normalizable.

Step 5: Truncate series

We need to truncate the sum so that $H(\xi)$ would at most be a polynomial of some finite degree n, and then $\psi(\xi) = H(\xi)e^{-\frac{1}{2}\xi^2}$ would be finite, since $e^{-\frac{1}{2}\xi^2}$ decays much stronger than any polynomial of finite degree at large ξ. We truncate the series at n by setting $a_{n+2} = 0$ while $a_n \neq 0$, which is achieved when $2n + 1 - K = 0$, or using the definition of K,

$$E_n = (n + \frac{1}{2})\hbar\omega, \tag{2.7.17}$$

where we could choose to truncate the series at $n = 0$, $n = 1$, $n = 2$, etc, which produce polynomials of the same degree. These energies are the energy levels of the simple harmonic oscillator, and the corresponding polynomials are the Hermite polynomials (up to irrelevant overall constant factors).

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